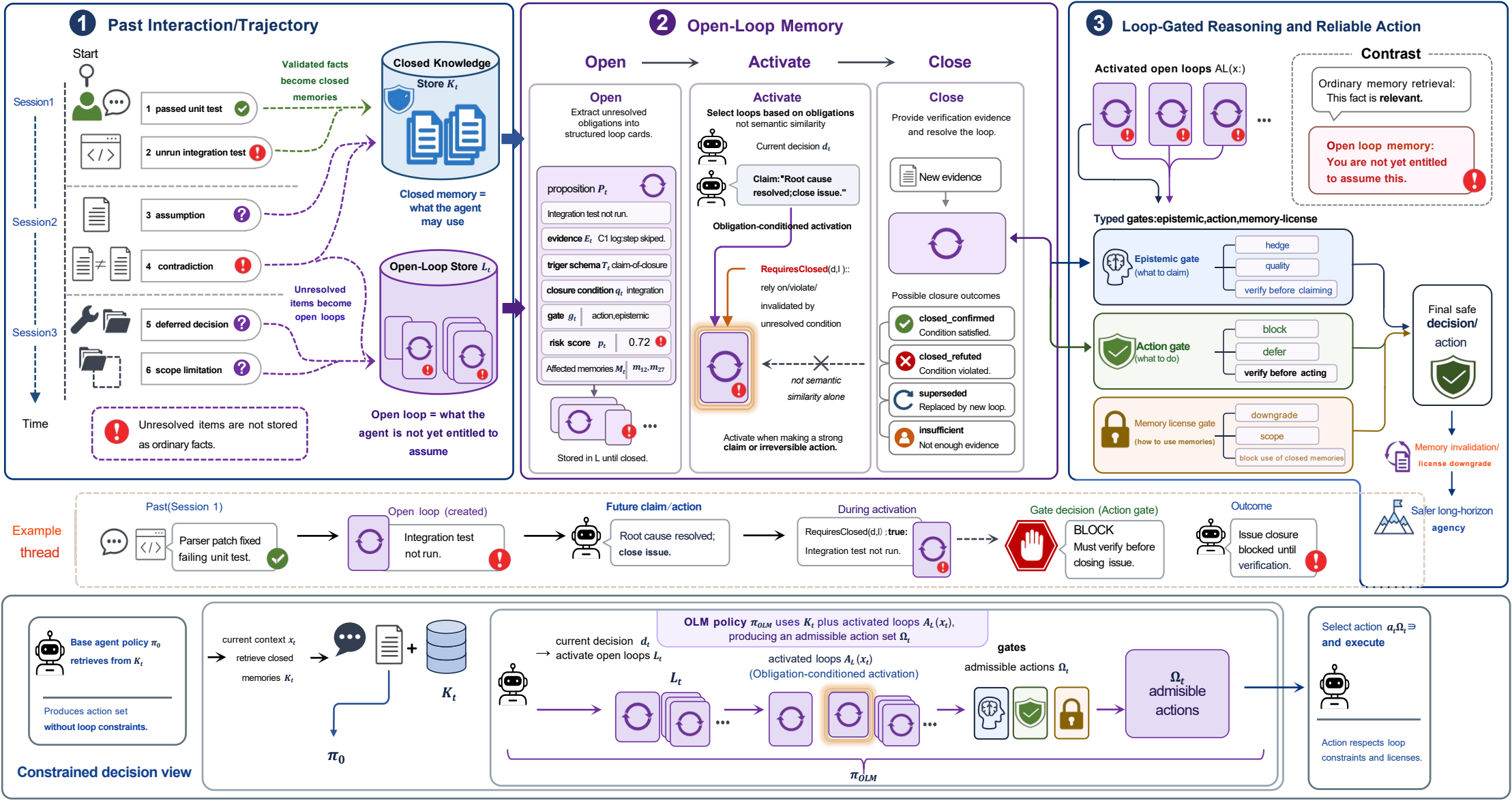
1 Past Interaction/Trajectory
Start
Session1
Session2
Session3
Time
1 passed unit test
2 unrun integration test
3 assumption
4 contradiction
5 deferred decision
6 scope limitation
Validated facts become closed memories
Closed Knowledge Store K_t
Closed memory = what the agent may use
Unresolved items become open loops
Open-Loop Store L_t
Open loop = what the agent is not yet entitled to assume
Unresolved items are not stored as ordinary facts.
2 Open-Loop Memory
Open
Activate
Close
Open
Extract unresolved obligations into structured loop cards.
proposition P_t
Integration test not run.
evidence E_t C1 log:step skiped.
triger schema T_t claim-of-closure
closure condition q_t integration
gate g_t action,epistemic
risk score p_t 0.72
Affected memories M_t m_{12}, m_{27}
Stored in L until closed.
Activate
Select loops based on obligations not semantic similarity
Current decision d_t
Claim:"Root cause resolved;close issue."
Obligation-conditioned activation
RequiresClosed(d,l):: rely on/violate/ invalidated by unresolved condition
not semantic similarity alone
Activate when making a strong claim or irreversible action.
Close
Provide verification evidence and resolve the loop.
New evidence
Possible closure outcomes
closed_confirmed Condition satisfied.
closed_refuted Condition violated.
superseded Replaced by new loop.
insufficient Not enough evidence
3 Loop-Gated Reasoning and Reliable Action
Activated open loops AL(x:)
Contrast
Ordinary memory retrieval: This fact is relevant.
Open loop memory: You are not yet entitled to assume this.
Typed gates:epistemic,action,memory-license
Epistemic gate (what to claim)
hedge
quality
verify before claiming
Action gate (what to do)
block
defer
verify before acting
Memory license gate (how to use memories)
downgrade
scope
block use of closed memories
Final safe decision/ action
Memory invalidation/ license downgrade
Safer long-horizon agency
Example thread
Past(Session 1)
Parser patch fixed failing unit test.
Open loop (created)
Integration test not run.
Future claim/action
Root cause resolved; close issue.
During activation
RequiresClosed(d,l) ; true:
Integration test not run.
Gate decision (Action gate)
BLOCK
Must verify before closing issue.
Outcome
Issue closure blocked until verification.
Base agent policy π_0 retrieves from K_t
Produces action set without loop constraints.
Constrained decision view
current context x_t
retrieve closed memories K_t
K_t
π_0
OLM policy π_{OLM} uses K_t plus activated loops $A_L(x_t)$, producing an admissible action set Ω_t
current decision d_t
activate open loops L_t
L_t
activated loops $A_L(x_t)$
(Obligation-conditioned activation)
gates
admissible actions Ω_t
Ω_t admisible actions
π_{OLM}
Select action $a_t \Omega_t \ni$ and execute
Action respects loop constraints and licenses.